Executive report: where is the advertising budget lost?

An investigation into 200,000 digital marketing campaigns (2021)

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Executive summary

We looked for the channel, audience or time of year that made advertising spend perform better. None did. The money is lost somewhere else: in specific campaigns that barely convert, spread across every channel. Almost 1 in 10 campaigns takes 10.9% of the budget (272.1 of 2,500.9 million USD) and brings only 1.3% of the conversions.

Three key messages

1. **Channel, audience, city, language, spend, duration and timing are not levers.** Every comparison was checked with statistical tests and none shows a real difference. Spending more does not even bring more conversions.
2. **The real lever is the campaigns that do not convert.** No targeting or timing decision sets them apart. Their conversion rate gives them away and, to a lesser extent, so does their budget: since spending more does not bring more conversions, the campaigns with the biggest budgets end up expensive more often.
3. **Better data is needed** to decide where, when and for how long to invest: retention, a consistent ROI, real dates and the daily evolution of each campaign.

Key figures

Indicator	Value
Campaigns analysed	200,000 (year 2021)
Median cost per conversion	334 USD
Gap between the best and worst channel	under 7 USD (2%), not significant
High-cost campaigns	9.2% of campaigns, 10.9% of budget, 1.3% of conversions
Cost per conversion, high-cost campaigns vs the rest	2,336 vs 256.5 USD
High-cost campaigns caught by the conversion ⩽ 2% rule (on the final result)	58.4%

Indicator	Value
High-cost campaigns among the 20% that spend least vs the 20% that spend most	2.9% vs 16.0%

Recommendations

1. Review the campaigns that end with a conversion rate of 2% or less, and check with daily data from what point in a campaign that rule can be applied.
2. Justify big budgets with the conversions they are expected to add, and review them first.
3. Do not move budget between channels, audiences or cities based on this data.
4. Measure the retention of acquired users, a consistently calculated ROI, the real dates of each campaign and its day-by-day results.

🎯 The question: where do we put the money?

A content brand (a streaming platform, a video game or an app) invests in advertising every year to acquire users. In the data analysed, that investment adds up to 2,500.9 million USD spread over 200,000 campaigns in 2021. The CEO's question is simple: **where does each dollar work hardest?**

Each campaign leaves a complete trail: the channel it ran on (Email, Facebook, Google Ads, Instagram, Website or YouTube), who it targeted, in which city and language, how long it lasted, how much it cost and what it achieved: impressions, clicks, conversions, ROI and engagement.

To answer, we followed the money as in an investigation. Any marketing team has a list of usual suspects when one campaign works better than another:

1. **The channel:** are some platforms cheaper at acquiring customers?
2. **The audience:** do certain segments, ages, cities or languages respond better?
3. **The combination:** is there a winning formula of channel, campaign type and audience?
4. **Budget, duration and timing:** does spending more, running longer or launching at the right moment improve results?

The yardstick is **cost per conversion**: what it costs to acquire each customer. For loyalty we use engagement, because the data has no retention. Each suspect went through a statistical test, so as not to mistake chance for a real difference.

🔍 First suspect: the channel

It is the first place any marketing director looks: is it cheaper to acquire customers on YouTube than on Facebook? The answer is no. The six channels acquire customers at practically the same cost, between 330.5 USD (Website) and 337.3 USD (Instagram). Click-through rate is around 10% and conversion rate around 8% in all of them.

The six channels acquire customers at almost the same cost

Median cost per conversion by channel (USD)



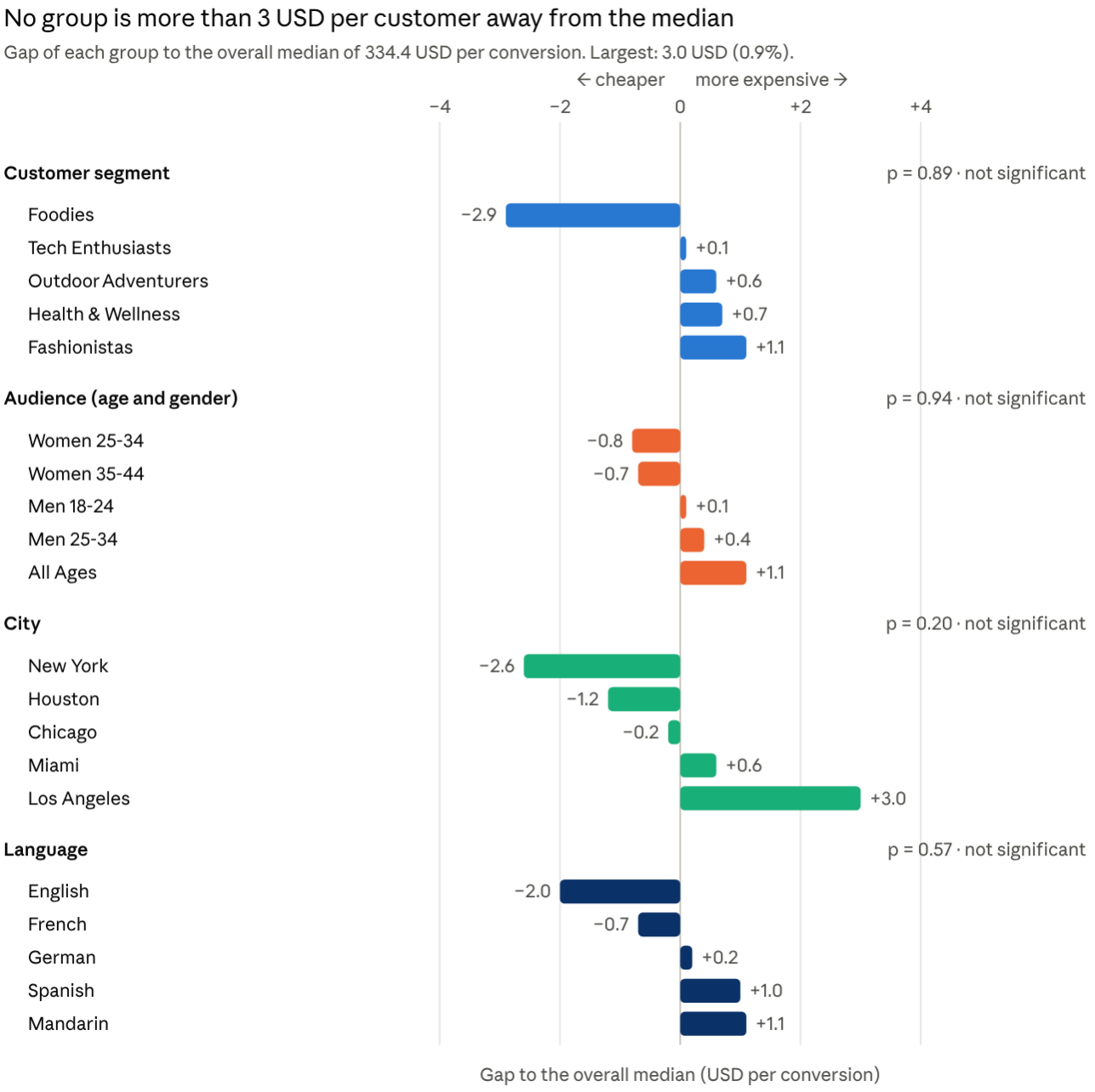
Gap between the cheapest and the most expensive channel: 6.8 USD (not significant, p between 0.17 and 0.93)

Project notebook, section 4.1 · 200,000 campaigns from 2021

A 7 USD gap might look like a lead, but the statistical test (Kruskal-Wallis) says it is noise. None of the seven metrics differs between channels: all p-values are between 0.17 and 0.93, far above the 0.05 threshold. **The channel has an alibi.**

👥 Second suspect: who we target

If the channel explains nothing, perhaps the audience does. We reviewed four dimensions: customer segment, audience by age and gender, city and language. The twenty groups pay between 331.5 and 337.4 USD per customer.



Project notebook, section 4.3 · 20 groups of about 40,000 campaigns each

Of 20 tests, only one fell below 0.05: ROI by audience. It looked like a lead, but when running so many comparisons it is normal for one to come out “significant” by pure chance. The Bonferroni correction confirms it: it is not real. **The audience has an alibi too.**

🎲 Third suspect: the winning formula

One possibility remained: that no factor works on its own, but a specific combination does. Crossing channel, campaign type and audience gives 150 combinations. Sorted in a table (below, the first five and the last five of the 150), a winner finally seems to appear: Website | Influencer | Men 25-34, at 310.6 USD per customer, against 366.4 USD for YouTube | Search | Men 18-24. That is a 56 USD gap.

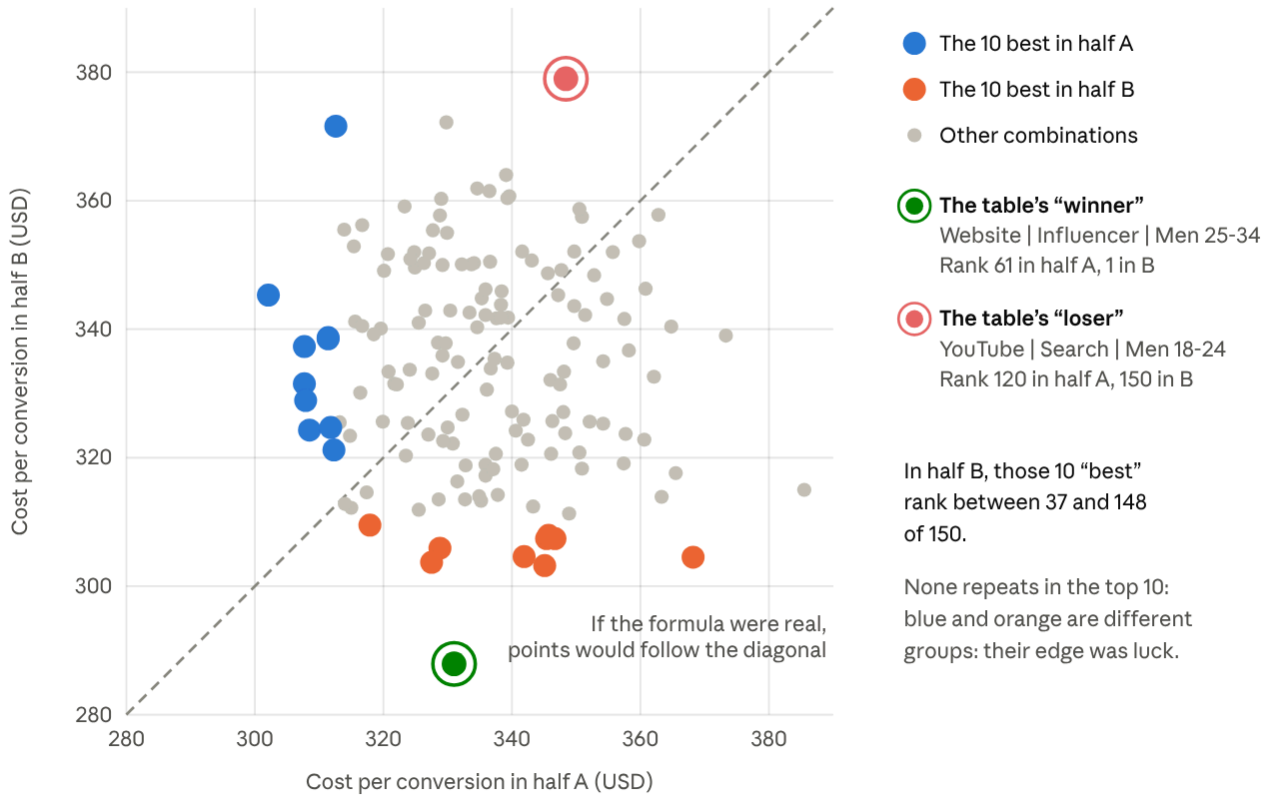
Rank	Channel	Campaign type	Audience	Cost per conversion (USD)
1	Website	Influencer	Men 25-34	310.6
2	Website	Search	All Ages	313.2
3	Instagram	Social Media	Men 18-24	313.4
4	Facebook	Email	Men 18-24	314.4
5	Facebook	Social Media	Men 18-24	315.2
146	Google Ads	Social Media	Men 18-24	353.0
147	Google Ads	Email	All Ages	355.8
148	Instagram	Display	Men 25-34	356.4
149	Facebook	Influencer	Women 35-44	360.2
150	YouTube	Search	Men 18-24	366.4

At first sight, the table invites moving budget towards the top rows. But there is a warning sign: the Men 18-24 audience appears three times among the best and twice among the worst.

To check it, we split the campaigns into two random halves and ranked each half separately. If the winning formula were real, it would win in both. It did not: **none of the 10 best combinations in one half appears among the 10 best in the other**. In the chart, the table's "winner" and "loser" are circled: the first is the best in half B, but only 61st in half A; the second is last in half B, but 120th in half A. The winner was a mirage, a product of chance when slicing the data into small groups.

The “winning” combinations do not repeat: the ranking is chance

Median cost per conversion of the 150 combinations in two random halves of the data (USD)



Project notebook, section 4.6 · 150 combinations, random halves with seed 42

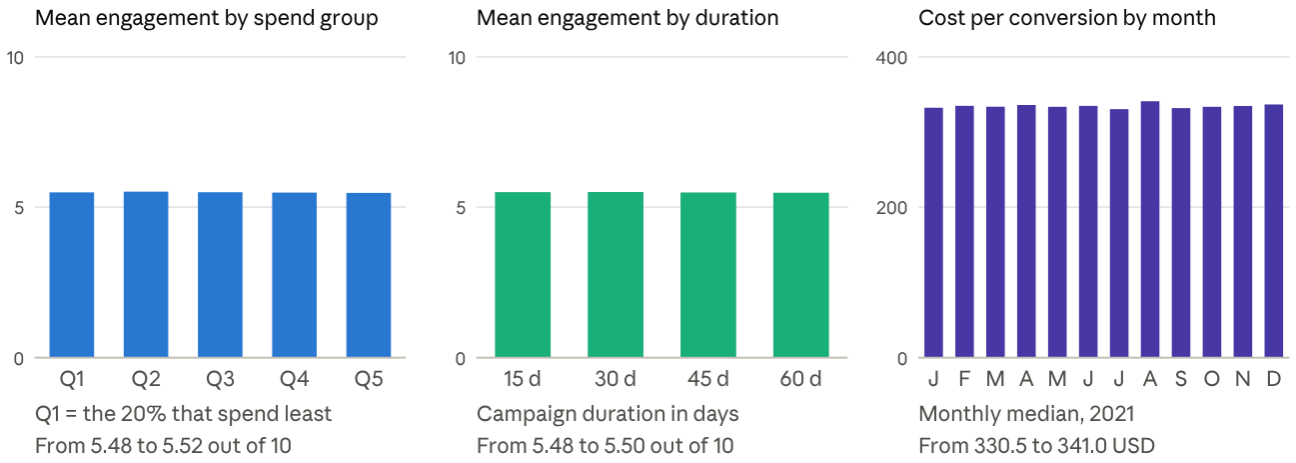
⏱️ Fourth suspect: budget, duration and timing

The last suspect is how the money is invested: spending more, running longer or choosing the right moment. Here we also look at engagement, the closest measure of the loyalty of acquired users.

- **More budget or more reach.** We split the campaigns into five equal groups by spend and by impressions. Mean engagement stays between 5.48 and 5.52 out of 10 in all of them, for 5,000 USD and 20,000 USD campaigns alike ($p = 0.337$ and 0.438). Conversions do not rise either: about 36 per campaign (median) in every spend group.
- **Longer duration.** At 15, 30, 45 or 60 days, median ROI is between 5.00 and 5.02 and cost per conversion between 332.9 and 335.3 USD. A 60-day campaign even spends the same as a 15-day one (about 12,500 USD median).
- **Time of year.** A 30-day moving average of cost per conversion only moves between 327.9 and 341.4 USD over 2021, with no peaks in summer or at Christmas. Campaigns launched on one of the 11 federal holidays, obtained from a public API, perform like the rest ($p = 0.688$).

Neither budget, nor duration, nor month changes the results

Each panel keeps its full scale, from 0

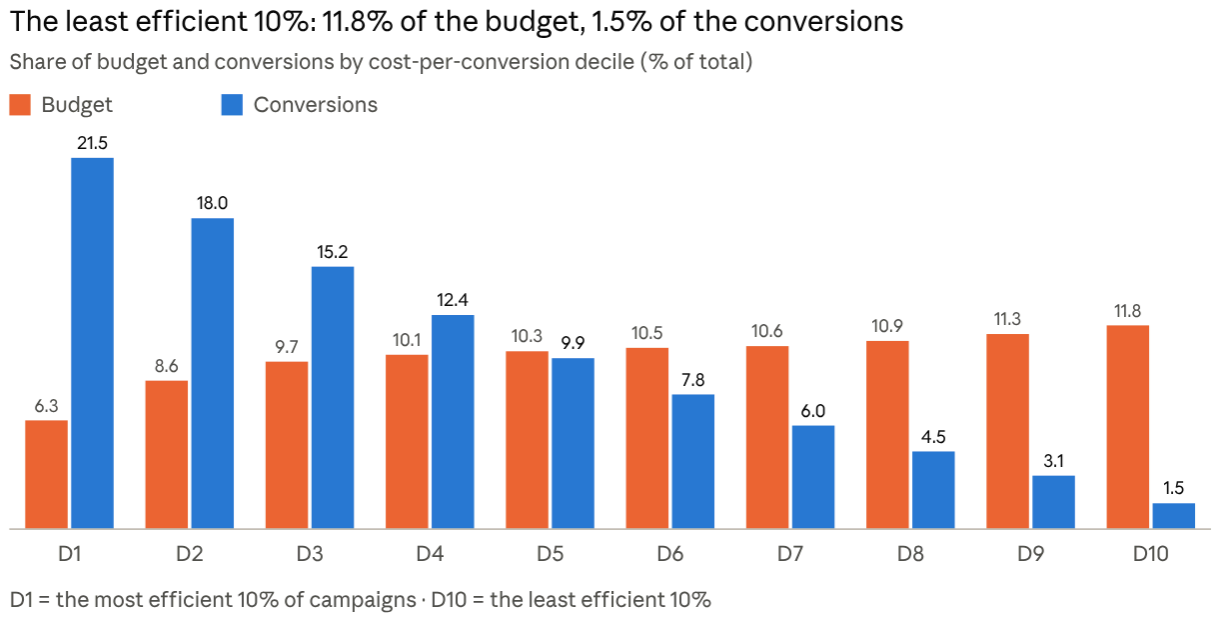


Project notebook, sections 4.2, 4.4 and 4.5 · 200,000 campaigns from 2021

Four suspects, four alibis. For a content brand, this means something uncomfortable: there is no evidence that bigger, longer or better-timed campaigns acquire more engaged users. If the problem is not in how the campaign is set up, where is it?

💡 The twist: the leak is inside every channel

The decisive clue was not in the campaigns' characteristics but in their results. Sorting the 200,000 campaigns from most to least efficient, the leak appears: **the least efficient 10% of campaigns take 11.8% of the budget and bring only 1.5% of the conversions**. At the other end, the most efficient 10% use 6.3% of the budget and bring 21.5% of the conversions.



Project notebook, section 4.6 · 200,000 campaigns from 2021

High-cost campaigns. The analysis flagged 9.2% of campaigns as high-cost: those that pay far more than usual per customer. They take 272.1 million USD (10.9% of the budget) and get 1.3% of the conversions. They pay 2,336 USD per conversion, against 256.5 USD for the rest.

The main problem is that they convert little. Compared with a normal campaign, they get less than half the clicks (232 vs 582) and a conversion rate of 2% instead of 9%. They also spend somewhat more: 15,444 vs 12,156 USD median. And here is the key to the whole investigation: **they are spread evenly across the six channels**. That is why no suspect looked guilty: the leak is not in one channel, it is inside all of them.

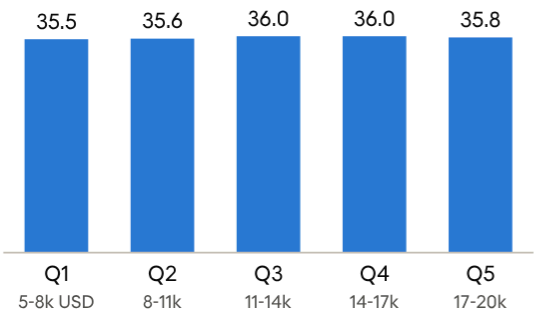
It is not just the channel. We checked 11 characteristics that are decided before a campaign is launched: channel, campaign type, audience, segment, city, language, company, duration, month, weekday and holiday. In every group, between 8.9% and 9.6% of campaigns are high-cost, and no test shows a real difference (p between 0.055 and 0.790).

The only pre-launch clue: the budget. There is one exception. Spending more does not bring more conversions: about 36 per campaign (median) in every spend group. Since cost per conversion is spend divided by conversions, more budget with the same conversions means more expensive customers. That is why the share of high-cost campaigns rises from 2.9% among the 20% of campaigns with the smallest budget (5,000 to 7,999 USD) to 16.0% among the 20% with the largest (17,014 to 20,000 USD): 5.5 times more.

More budget, same conversions: big campaigns end up expensive more often

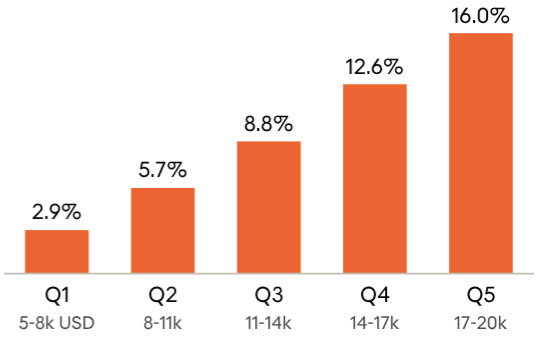
Campaigns split into five equal groups by budget (Q1 = the 20% that spend least)

Conversions per campaign (median)



Scale from 0 to 50: the bars are almost equal

High-cost campaigns (% of the group)



5.5 times more in Q5 than in Q1

Project notebook, section 4.6 · 200,000 campaigns in five groups of 40,000

A caveat. In real campaigns, more budget usually buys more impressions and clicks. Not in this dataset: spend is not related to any result. What the data does show is that here the extra budget did not buy extra conversions, and that has to be checked with real data.

What-if scenario. Had that budget converted at the average cost of the other campaigns, it would have brought about 944,000 more conversions, 10.7% more in total. This is an upper bound: it assumes the money could be moved without losing efficiency, which rarely happens.

🚨 The solution: the signal that gives failing campaigns away

Neither the channel, nor the audience, nor the duration, nor the timing tells in advance which campaign will turn out expensive, and the budget is only a weak clue (the table shows it). But its result gives it away: the conversion rate. A conversion rate of 2% or less flags 10.7% of campaigns and catches 58.4% of the high-cost ones; half of the flagged campaigns are high-cost.

Alert rule	Campaigns flagged	High-cost campaigns caught	Flagged that are high-cost
Conversion ⩽ 1%	3.5%	29.4%	77.2%
Conversion ⩽ 2%	10.7%	58.4%	50.3%
Conversion ⩽ 3%	17.8%	73.9%	38.3%

Alert rule	Campaigns flagged	High-cost campaigns caught	Flagged that are high-cost
Spend in the top 20% (known before launch)	20.0%	34.8%	16.0%
Conversion ⩽ 2% and spend in the top 20%	2.2%	17.2%	73.6%

A stricter threshold reviews fewer campaigns and is right more often; a wider one catches more, at the cost of reviewing campaigns that were fine. Budget alone is right in only 16.0% of the campaigns it flags; combined with conversion it is right in 73.6%, but catches few (17.2%).

What we do not know yet: when the signal appears. The dataset only stores the final result of each campaign, not its day-by-day evolution. So it cannot tell whether a campaign that will end up expensive is already recognisable in its first week or only at the end. Using conversion as an early warning is a hypothesis, not a finding.

Proposal to test it: a pilot that records each campaign's conversion rate at the end of its first week and compares it with its final result. If the campaigns that end at 2% or less are already below that threshold in their first week, the rule can be used to stop campaigns in time. If not, it will help decide which ones not to repeat.

🧪 What the data cannot tell

Every investigation has limits, and this one too. The data is clean, but not real: everything suggests it was generated at random, column by column. That explains why so many suspects had an alibi, and it means the figures must be read as relative differences, not as expected market costs.

What was fixed or added

- Spend, duration and date came as text and were converted to numbers and dates.
- Two columns had no definition. Conversion rate was assumed to be conversions over clicks, and acquisition cost the total campaign spend.
- High-cost campaigns were flagged, not removed, because they are exactly the waste this report looks for.
- The data was enriched with the 2021 US federal holidays, downloaded from the public Nager.Date API.
- There were no missing values, duplicates or misspelled labels.

What this dataset cannot measure

- **Independent columns.** Spend is not related to impressions, clicks or conversions (correlation of 0.00). Combinations with no business sense exist, such as 6,589 email campaigns run on Facebook.
- **Unrealistic scale.** The median click-through rate is 10% and the cost per thousand impressions 2,270 USD, far above usual advertising values.
- **Inconsistent ROI.** Campaigns that pay about 7 times more per customer have the same median ROI as the rest (5.00 vs 5.01).
- **Dates are not real.** The date follows the campaign number in a fixed 365-day rotation, so seasonality cannot be measured.
- **No retention data.** Engagement is the closest measure, and it responds to no campaign decision.
- **No evolution during the campaign.** Each campaign has a single row with its final result, so it is impossible to know at what point a campaign starts to show it is going badly.

🏁 Conclusions: from investigation to action

The investigation started with a question about where to invest and ends with a more useful one: **which campaigns do not convert and how to recognise them**. Channel, audience and timing do not decide performance; each campaign's result does, and the conversion rate makes it possible to recognise the ones that fail.

1. **Use conversion as an alert signal.** Review the campaigns that end at a conversion rate of 2% or less, and validate with a pilot of daily data from what point in a campaign the rule can be applied. The threshold can be adjusted to the team's capacity: 1% reviews fewer campaigns and is right in 77.2% of cases.
2. **Justify big budgets.** In this data, the 20% of campaigns with the largest budget get the same conversions as the 20% with the smallest (35.8 vs 35.5 median), and 16.0% of them end up high-cost, against 2.9%. Big campaigns should be reviewed first, and every budget increase justified with the conversions it is expected to add. In real campaigns more budget usually buys more reach, so this must be confirmed with real data before setting spending limits.
3. **Do not move budget between channels or audiences based on this data.** The differences are noise and would not repeat. Channel and audience choices should follow brand and market strategy.
4. **Measure what matters for a content brand.** The value of a campaign lies in how long acquired users stay. The next analysis needs day-7 and day-30 retention, an ROI calculated from spend and revenue, real dates, daily results for each campaign and A/B tests to measure the real effect of each channel.

5. **Read the figures as relative.** The absolute values in this dataset do not work as market benchmarks.

Methodology note. Groups were compared with medians, because cost per conversion is strongly skewed, and with the Kruskal-Wallis test and its effect size. Relationships between variables were measured with Spearman correlation, and the yearly trend with a 30-day moving average. The Bonferroni correction was applied when running many comparisons, and the combinations were validated by splitting the data into two halves. Cleaning and analysis are in the repository notebook, with functions covered by tests and a reproducible pipeline.